

AI Capability	Application	TensorFlow	MXNet	Caffe2	Caffe
Image Recognition	Classify image(s) with high accuracy	ResNet101 (RT) InceptionV3 (RT)	ResNet101 (RT) InceptionV3 (RT) InceptionV4 (RT) Xception (RT) SqueezeNet (RT) DenseNet (RT)	ResNet101 (RT) InceptionV3 (RT) SqueezeNet (RT) VGG16, VGG19	ResNet101 (RT) InceptionV3 (RT)
Object Detection & Localization	Locate and classify object(s) in image	SSD+VGG16 (RT)	SSD (RT)		SSD+VGG16 (RT) Faster-RCNN (RT) Fast-RCNN (RT)
Speech Recognition	Convert speech to text		Deep Speech 2 (RT)		
Language Translation	Translation from one language to another	Seq2Seq (RT)	Seq2Seq (RT) Transformer (RT)		
Recommender Systems	Predicts the rating or preference a user would give an item		Coxie & Deep (RT)		
Generative Adversarial Networks	Neural networks that generate data mimicking some distribution		DCGAN (RT)		
Reinforcement Learning	The use of actions and rewards to learn how to behave in an environment		A3C (RT)		

CPU-optimised deep learning topologies for several popular frameworks

plays an active role in bringing up new additions and updates to the software with quick error resolution and query matching.

There is also the Intel® Data Analytics Acceleration Library which has a performance library of highly optimised algorithms for all the major processes in the analytic data stages.

It is designed to support a majority of popular data science platforms like Hadoop, Spark, R, Python and OpenCV. All available for free through community licensing.

WHY INTEL FRAMEWORKS RANK WELL

With AI becoming a mainstream technology, many technology companies have been developing specialised hardware and software for AI applications. Intel is no different in this regards as it boasts of a huge portfolio of hardware and software for AI training and inference. A majority of reviews and advice columns would suggest that the Intel® Xeon™ series is perhaps the most compatible CPU family that could work for all sorts of deep learning projects.

To understand how well software that is optimised for Intel hardware can make a difference, let's take the example of Caffe*, a deep learning framework developed by the kind folks over at Berkeley Vision and Learning Center (BVLC) and community contributors. There's the vanilla version of